

1. What are Lean and Agile?

Lean

Lean means doing more with less by removing activities that do not create value for the customer.

Think of it as asking:

“Are we spending time, money, or effort on something that the customer does not actually need?”

For example, an e-commerce company may have a process:

Customer orders → Staff print order → Staff re-enter order into another system → Manager approves → Warehouse receives order → Staff pick item

If the order information can automatically move from the website to the warehouse system, manually re-entering the order is **waste**.

Lean would try to eliminate that unnecessary step.

Agile

Agile means working in small, fast cycles, getting feedback, and adapting quickly when requirements change.

Instead of:

Plan everything for 12 months → Build everything → Launch → Discover customers don't like it

Agile uses:

Plan → Build a small part → Test → Get feedback → Improve → Build the next part

Simple comparison

Lean	Agile
Focuses on reducing waste	Focuses on responding to change
Improves efficiency	Improves adaptability
Removes unnecessary activities	Delivers in small iterations
Asks "What can we eliminate?"	Asks "What should we improve next?"
Example: automate manual data entry	Example: release a new app feature every 2 weeks

Together: Lean + Agile helps organisations **deliver value faster, with less waste, while adapting to customer needs.**

2. The 8 Digital Wastes

The slide mentions digital wastes such as **defects, overprocessing, waiting, unused talent and unnecessary data movement.**

A useful way for students to understand digital waste is:

Digital waste is time, money, data, effort or technology being used without creating useful value for the customer or organisation.

Let's look at the major types.

Waste 1: Defects

Meaning

A **defect** occurs when a digital product, service or process does not work correctly.

Real-world example: Online shopping

Imagine an online fashion store.

A customer:

1. Selects a shirt.
2. Adds it to cart.
3. Pays \$50.
4. Receives an error message.
5. Payment was actually successful.
6. The order is not recorded.

The company now needs to:

- investigate the transaction
- refund the customer
- answer customer complaints
- correct the database
- potentially lose the customer

All of this creates waste.

Lean solution

Use:

- automated testing
- continuous integration
- monitoring
- error logging
- quality checks

Key idea:

Prevent defects early rather than fixing expensive problems later.

3. Waste 2: Overprocessing

Meaning

Overprocessing means doing more work than necessary to produce the required result.

Real-world example: School administration

Suppose a student applies for a course.

The student submits:

Name + NRIC/ID + email + qualification

But the administrative staff then:

1. Enter the information into Excel.
2. Enter it again into the Student Management System.
3. Print the application.
4. Scan the application.
5. Upload the scanned copy.
6. Email the information to another department.

If the same information could automatically flow into the student system, many of these steps are unnecessary.

Lean solution

Use:

Online application → Automated database → Student record

instead of:

Online application → Excel → Print → Scan → Email → Manual entry

Student takeaway

If the same information is repeatedly processed, copied or checked without adding value, it may be overprocessing.

4. Waste 3: Waiting Time

Meaning

Waiting occurs when people, systems or processes cannot continue because they are waiting for something else.

Real-world example: Food delivery

Imagine a customer orders food through a delivery app.

The restaurant receives the order but:

waits 15 minutes before accepting it.

Then the driver arrives but:

waits another 20 minutes for the food.

Then the customer waits:

another 15 minutes for delivery.

The customer may become dissatisfied.

Digital solution

The platform can provide:

- real-time order status
- automatic restaurant notifications
- driver allocation
- estimated preparation time
- real-time delivery tracking

For example:

Order placed → Restaurant notified → Food preparation → Driver assigned → Pickup → Delivery

Key idea

Lean tries to reduce unnecessary waiting between activities.

5. Waste 4: Unused Talent

Meaning

This happens when employees have useful skills but the organisation does not use their knowledge effectively.

Real-world example

A company employs a software developer who is very good at automation.

However, the developer spends most of the week manually preparing Excel reports.

At the same time, another employee spends hours manually copying data.

This is a waste of **human potential**.

Better approach

The developer could:

- automate the report
- create a dashboard
- build a data pipeline
- train employees to use the system

The organisation benefits from the employee's actual expertise.

Key idea

Lean is not only about reducing technology costs. It is also about making good use of people's skills.

6. Waste 5: Unnecessary Data Movement

Meaning

Unnecessary data movement occurs when information is repeatedly transferred between systems or people when it could be integrated automatically.

Real-world example: E-commerce

Imagine:

Shopify → Excel → Accounting System → Warehouse System

Staff manually download and upload files every day.

This creates:

- time consumption
- duplicate work
- data-entry errors
- delays

Better approach

Use APIs/integrations:

Shopify → API → Accounting + Inventory + Warehouse

Now the data moves automatically.

Example

When a customer buys:

iPhone case × 1

The system automatically:

- records the sale
- reduces inventory
- generates an order
- updates accounting
- informs warehouse staff

Key idea

Automate information flow whenever possible.

7. Iterative Sprints

What is a Sprint?

A Sprint is a short, fixed period in which a team works on a specific set of tasks and produces a usable result.

A common Agile Sprint is **1–4 weeks**.

The slide uses a **2-week execution cycle**.

Instead of building an entire system for six months, the team develops it gradually.

Traditional approach

Imagine a food delivery company wants to create a new app.

The team spends:

Month 1–2: Requirements

Month 3–4: Design

Month 5–6: Development

Month 7: Testing

Month 8: Launch

Only after eight months does the company discover that customers want a different feature.

That's expensive to change.

Agile approach

Sprint 1

Build:

Customer registration + login

Sprint 2

Build:

Restaurant search

Sprint 3

Build:

Shopping cart

Sprint 4

Build:

Payment

Sprint 5

Build:

Delivery tracking

After every Sprint:

Build → Test → Demonstrate → Get feedback → Improve

Real-world example: Netflix

A digital service like Netflix constantly improves its platform rather than waiting several years to release one huge redesign.

Teams can work on areas such as:

- search
- recommendations
- user interface
- streaming performance
- accessibility
- account management

Small improvements can be tested and refined based on usage and feedback.

Key idea

Don't try to build everything perfectly at once. Build, learn and improve.

8. Pull Systems

What is a Pull System?

A Pull System means work is triggered by actual demand rather than producing work simply because it is possible.

Think:

Customer demand pulls the work through the system.

Real-world example: Food delivery

Suppose a restaurant prepares 500 meals in advance even though only 100 customers are expected.

The restaurant has:

- wasted ingredients
- wasted labour
- excess food
- storage problems

A pull system works differently.

Customer places order

↓

Restaurant receives demand

↓

Food preparation begins

↓

Food delivered

The demand "pulls" the work.

9. Pull System in Digital Operations

Consider an online store.

A customer orders:

Wireless keyboard

The order triggers:

1. Inventory check
2. Warehouse picking
3. Packing
4. Shipping
5. Delivery tracking
6. Customer notification

The company doesn't unnecessarily process thousands of orders that nobody has placed.

Another example: Cloud computing

A website may automatically increase server capacity when traffic increases.

For example:

Normal traffic → 3 servers

High traffic → Automatically scale to 10 servers

Traffic decreases → Scale back to 3 servers

This is a demand-driven approach.

Key idea

Don't produce or process more than the customer demand requires.

10. Continuous Improvement – Kaizen

What is Kaizen?

Kaizen means continuous, incremental improvement.

Instead of asking:

"How can we completely redesign everything?"

Kaizen asks:

"What small improvement can we make today?"

Real-world example: Amazon warehouse

Imagine warehouse workers spend an average of 30 seconds searching for each product.

The company could continuously analyse the process and make small improvements:

Improvement 1

Move popular products closer to packing stations.

Improvement 2

Use barcode scanners.

Improvement 3

Introduce better warehouse layouts.

Improvement 4

Use predictive inventory data.

Improvement 5

Use automation/robots for repetitive movement.

Each improvement may seem small, but together they can significantly improve operational efficiency.

11. Kaizen in a School

This concept can also be applied to **Genetic Computer School**.

Imagine the student enrolment process currently requires:

Student submits form → Admin checks → Academic checks → Finance checks → Data entry → Email confirmation

The school could review the process every month.

Improvement 1

Create an online application form.

Improvement 2

Automatically validate required fields.

Improvement 3

Automatically notify the relevant department.

Improvement 4

Automatically generate the student record.

Improvement 5

Automatically send confirmation email.

Eventually:

Student applies → System validates → Departments notified → Student record created → Confirmation sent

The process becomes faster and requires less manual work.

12. Cross-Functional Teams

The slide also refers to **cross-functional teams**.

What does this mean?

Instead of having only programmers make decisions, an Agile team may include:

- Developer
- UI/UX designer
- Operations staff
- Marketing staff
- Business analyst
- Data analyst
- Customer support representative

Real-world example: E-commerce website

Suppose an online store has a high cart-abandonment rate.

A cross-functional team investigates:

Marketing

→ Are customers receiving the wrong traffic?

UX designer

→ Is the checkout confusing?

Developer

→ Is the website slow?

Data analyst

→ At which stage do customers leave?

Customer service

→ What complaints are customers making?

Together they can identify the problem faster.

13. Agile + Lean Together

These concepts work particularly well together.

Imagine an online food delivery company.

Problem

Customers complain:

"My order takes too long."

Lean approach

Identify waste:

- restaurant waiting time
- manual order processing
- inefficient driver allocation
- duplicate data entry

Agile approach

Create improvements through short Sprints.

Sprint 1: Improve restaurant notification

Sprint 2: Improve driver assignment

Sprint 3: Add real-time order tracking

Sprint 4: Improve delivery prediction

Sprint 5: Analyse customer feedback

Each Sprint produces improvements.

14. Complete Real-World Example

Consider **Amazon-style e-commerce operations**.

Customer

Searches for:

Running shoes

Digital platform

Recommendation system suggests products.

Customer orders

↓

Inventory system automatically checks stock.

↓

Warehouse receives the order.

↓

Worker/robot picks product.

↓

Product is packed.

↓

Courier receives delivery information.

↓

Customer receives tracking information.

↓

Customer receives product.

↓

Customer provides feedback.

The organisation can then analyse:

- delivery time
- customer satisfaction
- return rate
- inventory accuracy
- order errors

and identify improvements.

This represents the combination of:

Lean

→ Remove unnecessary steps.

Agile

→ Continuously improve the digital service.

Pull

→ Customer order triggers the process.

Automation

→ Reduce manual work.

Kaizen

→ Continuously improve the process.

Data

→ Use real-time information to make decisions.

15. Easy Way for Students to Remember

Students can remember the concepts using this formula:

LEAN = REMOVE WASTE

Ask:

"What are we doing that doesn't add value?"

AGILE = RESPOND QUICKLY

Ask:

"How can we deliver and improve something quickly?"

SPRINT = WORK IN SMALL CYCLES

Ask:

"What can we complete in the next 1–2 weeks?"

PULL = RESPOND TO REAL DEMAND

Ask:

"Is there actual customer demand for this?"

KAIZEN = KEEP IMPROVING

Ask:

"What small improvement can we make next?"